

# Swathi Shree Narashiman

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## EDUCATION

- **Indian Institute of Technology Madras** : B.Tech Electrical Engineering (Hons.) & Minor in Computing  
**CGPA:** 9.64/10.0 Nov '22 – Present
- **DAV Group of Schools, Vellore** - Science Stream (CBSE)  
**Class 12:** 498/500 **Class 10:** 493/500 Jun '20 – Jul '22

## SCHOLASTIC ACHIEVEMENTS

- Recipient of S. Subramaniam Award (1<sup>st</sup> year), M. Saradha & Sankariah Scholarship (2<sup>nd</sup> year) for securing **highest CGPA** in B.Tech IIT Madras.
- One among the **100 Opjems Scholars** in India for the year 2024 (issued by O.P. Jindal Group)
- Secured All India Rank **759** in JEE Advanced 2022 and All India Rank **606** in JEE Main 2022
- Was issued **KVPY Fellowship** by IISc Bangalore for securing All India Rank **475** in KVPY-SX 2022

## PUBLICATIONS

1. F. D'Angelo, O. K. Yüksel, **Swathi Shree Narashiman**, *et al.*, "Induction heads interpolate n-grams," in *Forty-third International Conference on Machine Learning*, 2026. [Online]. Available: <https://openreview.net/forum?id=BSY7jhBxM1>
2. H. Balasundaram, **Swathi Shree Narashiman**, P. Mathur, *et al.*, *Semantic smoothing for language models via distribution estimation and embeddings*, 2026. arXiv: [2605.07994](https://arxiv.org/abs/2605.07994) [cs.IT]. [Online]. Available: <https://arxiv.org/abs/2605.07994>
3. S. Hussein, J. Guan, **Swathi Shree Narashiman**, *et al.*, "3d shape reconstruction from autonomous driving radars," in *The Thirty Sixth British Machine Vision Conference*, 2025
4. D. Panda, **Swathi Shree Narashiman**, J. K. Joe, *et al.*, *Fine-grained control over music generation with activation steering*, 2025. arXiv: [2506.10225](https://arxiv.org/abs/2506.10225) [cs.SD]
5. **Swathi Shree Narashiman** and N. Chandrachoodan, *Alphazip: Neural network-enhanced lossless text compression*, 2024. arXiv: [2409.15046](https://arxiv.org/abs/2409.15046) [cs.IT]
6. **Swathi Shree Narashiman**, V. A, D. Joshi, *et al.*, *Chipelets on wheels: Review paper on holistic chipelet solutions for autonomous vehicles*, 2024. arXiv: [2406.00182](https://arxiv.org/abs/2406.00182) [cs.AR]

## RESEARCH & EXPERIENCE

- **Bachelor Thesis** Jan '26 – May '26  
*Mentor:* [Prof. Andrew Thangaraj](#), IIT Madras Chennai, India
  - Proposed embedding based **Semantic Smoothing for Distribution Estimation** in statistical language models
  - Established minimax-optimal guarantees for distribution estimation with side information.
  - Demonstrated consistent perplexity improvements over classical smoothing methods. Work submitted at ITW 2026.
- **Summer@EPFL Research Internship** May '25 – Jul '25  
*Mentor:* [Prof. Nicholas Flammarion](#), TML Lab, EPFL Lausanne, CH
  - Researched **Mechanistic Interpretability** in transformers, focusing on In-Context Learning and Induction Heads
  - Investigated theoretical guarantees and empirical results for **Markovian prediction** in Large Language Models
  - Established that transformers approximate finite-length Markov chains using weighted unigram and bigram counts
  - Conference paper submitted & **accepted to the International Conference on Machine Learning (ICML), 2026.**
- **Adversarial Attacks in Large Language Models** Sep '24 – Jun '25  
*Mentor:* [Prof. Nambi Seshadri](#), UC San Diego Remote
  - Investigated **error propagation** in neural networks via logit-space attacks on open-source LLMs
  - Analyzed the impact of injected noise and channel erasures on BLEU, Perplexity, and Semantic Similarity metrics
  - Verified that LLMs are robust to noise, while BLEU scores vary by  $\sim 30\%$ , preserving linguistic semantics [\[Report\]](#)
- **Music Interpretability via Activation Steering** Apr '24 – May '25  
*Club:* [AI Club](#), Center for Innovation, IIT Madras Chennai, India

- Used **steering** and **residual stream analysis** for controllable **fusion music generation** via generative models
- Enabled genre, instrument, and pitch transfer efficiently by training  $<0.005\%$  parameters on  $<1K$  samples
- Published results & findings on Fine-grained control over music generation on Arxiv [[Preprint](#)]
- **Summer@EPFL Research Internship** May '24 – Aug '24  
Lausanne, CH  
Mentor: [Prof. Haitham Al Hassanieh](#), SENS Lab, EPFL
  - Selected among the **top 1.6% globally** to conduct research on **3D reconstruction** from **autonomous driving radars**
  - Enhanced the reconstruction using PCN and KL-based knowledge transfer, achieving a Chamfer distance below 7
  - Conference paper submitted & **accepted to the British Machine Vision Conference (BMVC) 2025**, Sheffield, UK
- **Neural Network-Enhanced Lossless Text Compression** Aug '23 – Sep '24  
Chennai, India  
Mentor: [Prof. Nitin Chandrachoodan](#), IIT Madras
  - Leveraged LLMs in rank-based predictive compression, achieving  $4.5\times$  compression on the **Gutenberg corpus**.
  - Analyzed diverse neural architectures coupled with information-theoretic frameworks like Huffman, Gzip etc.
  - Conducted empirical evaluations on domain-specific LLMs and published findings on arXiv [[Preprint](#)]

## TECHNICAL PROJECTS

- **CS6170: Randomized Algorithms Course Project** Jan '25 – May '25  
Chennai, India  
Instructor: [Prof. Yadu Vasudev](#), IIT Madras | [Report](#) | [Video](#)
  - Explored **Online algorithms** for **AdWords** & **bipartite matching** under adversarial & stochastic models
  - Surveyed primal-dual & perturbation-based techniques including recent improvements over the classical  $1 - \frac{1}{e}$  bound
- **EE6180: Advanced Topics in AI Course Project** (with [Immerso.ai](#) startup) Jan '25 – May '25  
Chennai, India  
Instructor: [Prof. Pravin Nair](#), IIT Madras | [Report](#) | [Github](#)
  - Trained Stable Diffusion models via **Textual Inversion** ( $< 1M$  params) to learn movie characters from 5-10 images
  - Designed prompt templates and token strategies to improve realism and control in few-shot generation
- **EE6150: Stochastic Modeling Course Project** Jan '25 – May '25  
Chennai, India  
Instructor: [Prof. Avhishek Chatterjee](#), IIT Madras | [Report](#)
  - Studied Glauber dynamics in sparse Ising models with near-linear mixing via graph decomposition
  - Examined Log-Sobolev methods and structural partitioning for efficient sampling in community detection

## SKILLS

Python, C, C++, PyTorch, TensorFlow, LLMs, OpenCV, Git, Verilog, Xilinx Vivado, ARM, Arduino IDE, Altium Designer (PCB)

## RELEVANT COURSEWORK

- **Machine learning & AI** : Advanced Topics in AI; Stochastic Modeling & Theory of Queues; Natural Language Processing
- **Algorithms & Computing** : Randomized Algorithms; Quantum Algorithms; Computer Organization; Cryptography
- **Mathematics & Theoretical Foundations** : Linear Algebra; Convex Optimization; Incentive Centered Design (Game Theory); Probability; Control Systems; Information Theory; Multivariable Calculus; Series & Matrices

## TEACHING EXPERIENCE

- **Teaching Assistant for Computer Organization** Jul '25 – Nov '25  
Chennai, India  
Instructor: [Prof. Janakiraman Viraraghavan](#), IIT Madras
  - Undergraduate course on Computer Organization: Evaluated examination scripts & conducted tutorial sessions.
- **Avanti Mentor** Nov '22 – Jul '23  
Chennai, India  
Club: [Avanti Fellows](#), IIT Madras
  - Provided one-on-one mentorship for underprivileged students to assist them in IIT-JEE preparation

## POSITIONS OF RESPONSIBILITY

- **Project Lead of AI Club** Apr '24 – May '25  
Chennai, India  
[AI Club](#), Center for Innovation, IIT Madras
  - As a core team member, led a team of 40 enthusiasts in the development fusion music generation using AI
  - Presented at the **Open House** event, attracting 500+ attendees and multiple media outlets
- **Core member of EE Research Club** May '24 – present  
Chennai, India  
[EE Research Club](#), IIT Madras

- Leading 30-member EE Research Club focused on cultivating a research mindset and building a scientific community
- Ideated, organized, and hosted events to foster research engagement like socials, professor interviews, tech talks, etc.
- **Powertrain Engineer of Raftar Formula Racing** *Apr '23 – Mar'24*  
*Raftar Formula Racing, Center for Innovation, IIT Madras* *Chennai, India*
  - Designed and tested key EV circuitry in the Low Voltage Subsystem of IITM's Formula Student electric race car
  - Team secured **Runners-up** position at **Formula Bharat 2024** with an international-level vehicle build
- **Press Correspondent** *May '25 – present*  
*The Fifth Estate, IIT Madras* *Chennai, India*
  - Contributed articles to the official student press, covering diverse aspects of campus life and student affairs
- **Contingent** *Oct '23 – Dec'23*  
*Inter IIT Tech Meet 12.0* *Chennai, India*
  - Proposed a chiplet-based solution to Jaguar Land Rover's High Prep problem as institute representative
  - Published on arXiv a cost-efficient SoC-to-chiplet design optimizing die size, interconnects, and architecture [[Preprint](#)]